Ofqual QAN	**600/0406/X**
Quartz ID	**1156, 1157**
Qualification	**Level 4 Diploma in Computing, Level 4 Diploma in Computing (with Business Management)**

Assessment	**Computer Systems 2.0**
Assessment Type	**Global Examination**
Date	**Wednesday 5th March 2025**
Time	**2.00PM-5.00PM**

Examination Paper

Answer ALL questions.

Clearly cross out any surplus answers.

Time: 3 hours

The maximum mark for this paper is 100.

Any reference material brought into the examination room must be handed to the invigilator before the start of the examination.

Answer ALL questions
Do NOT copy the questions to your answer booklet

Marks

Question 1

For each answer, only provide the corresponding letter (e.g., i, ii, iii, or iv).

a) Which of the following is NOT a characteristic of modern computer hardware? **1**

i) CPU
ii) Memory
iii) Operating System
iv) Cache

b) Which component of a CPU is responsible for directing the flow of data and instructions? **1**

i) Control Unit
ii) Memory
iii) ALU
iv) Cache

c) Which of the following is a characteristic of quantum computing? **1**

i) Uses classical bits for processing
ii) Utilises qubits to represent and process information
iii) Operates on a binary logic framework
iv) Relies solely on traditional hardware components

d) Which of the following is used to connect a CPU to other components in a computer? **1**

i) Cache
ii) Memory
iii) Bus
iv) Control Unit

e) In which numbering system is the digit 'F' used to represent a value? **1**

i) Hexadecimal
ii) ASCII
iii) Octal
iv) Extended ASCII

f) Which protocol is responsible for ensuring error-free transmission in data networks? **1**

i) HTTP
ii) SMTP
iii) FTP
iv) TCP

g) What is the primary challenge for quantum computing today? **1**

i) Lack of enough qubits to perform complex tasks
ii) High power consumption
iii) Difficulty in interfacing with classical systems
iv) Limited quantum algorithms available

h) The primary function of memory in a computer system is to: **1**

i) Process data
ii) Communicate with external devices
iii) Control data flow
iv) Store data temporarily or permanently

© NCC Education Limited 2025

i) Which of the following is **TRUE** about a computer bus? **1**

i) It is used only for transferring data between memory and the CPU

ii) It connects the CPU to external devices like printers and monitors

iii) It is a collection of wires that transfers data, addresses and control signals between various components of a computer system

iv) It can only transfer data in one direction at a time

j) Which of the following is a core characteristic of a distributed system? **1**

i) Centralised control

ii) A single computer manages all tasks

iii) Multiple computers work together as a single system

iv) All resources located in a single physical location

Total 10 Marks

© NCC Education Limited 2024

Question 2

a) Outline TWO (2) additional tasks carried out by the OS. Provide a description for each. **6**

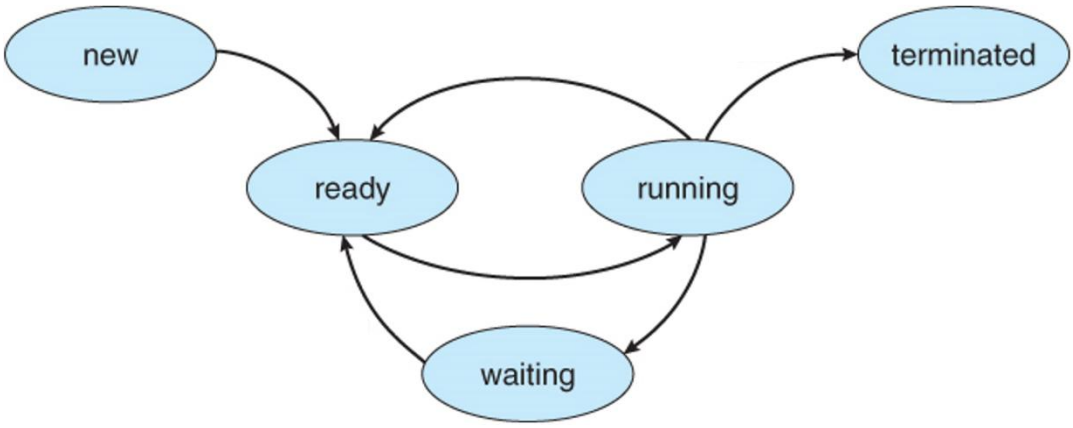
b) Describe the difference between pre-emptive and non-pre-emptive scheduling in an operating system. **2**

c) Other than fault tolerance and reliability, state and describe one additional distinct characteristic of a RTOS's operation concerning task and resource management. **2**

Total 10 Marks

Question 3

a) Using the following diagram, explain each of the FIVE (5) different states a process can be in. **5**



b) Timesharing is a technique associated with the management of a multi-user system. **4**

Describe TWO (2) distinct advantages and TWO (2) distinct disadvantages of timesharing.

c) State where both program data and instructions are stored in a Von Neumann architecture. **1**

Total 10 Marks

© NCC Education Limited 2025

Question 4

a) Convert the following decimal value to binary: **87**. Provide your answer in 8 bits. Show your workings. **3**

b) Subtract hexadecimal 58 from A2 in hexadecimal. Express the result in hexadecimal. **4**

Show your workings.

c) Add the unsigned binary numbers 10101 and 00100. Give the result in binary. **3**

Show your workings.

Total 10 Marks

Question 5

a) Name the THREE (3) main stages of a CPU's instruction cycle. **3**

b) Explain the purpose of an operand in a CPU instruction? **2**

c) Explain the purpose of an opcode in a CPU instruction. Give ONE (1) example. **3**

d) Before the CPU processes any data, it is recommended that data is checked via a parity checker. Describe how a parity checker works. **2**

Total 10 Marks

Question 6

a) Explain FOUR (4) requirements to ensure successful global network communication. **4**

b) State the difference between HTTP and HTTPS. **1**

c) In the OSI 7-layer reference model of network communications, what layer does logical addressing take place within? **1**

d) Bus and Star represent two different types of network configurations. **4**

For each network, state their fault tolerant and performance characteristics.

Total 10 Marks

© NCC Education Limited 2025

Question 7

a) A logic gate has TWO (2) inputs and one output. The only time its output is TRUE is when both inputs are TRUE. Identify the type of this gate. **2**

What gate will it be called if both inputs are negated?

b) An electrical system has TWO (2) inputs that control the flow of power. Power stops only when both inputs are closed. What logic does this represent? **1**

c) What logic name is given to the system where electricity stops only when both inputs are open? **1**

d) Complete the following table for the logic function: **A.(B+C)** **6**

A	***B***	***C***	B+C	A.(B+C)
0	*0*	*0*	0	0
0	*0*	*1*	1	
0	*1*	*0*		0
0	*1*	*1*	1	
1	*0*	*0*	0	0
1	*0*	*1*	1	
1	*1*	*0*	1	1
1	*1*	*1*		

Total 10 Marks

Question 8

a) Explain how system software differs from application software. **3**

b) A system software that is essential for producing application software is the language compiler. Explain the main function of a compiler. **2**

c) When developers create software, they sometimes use virtual machine. Explain what a virtual machine is and what it is used for. **5**

Total 10 Marks

Question 9

a) There are two types of instruction set computers. Provide the full name of each type and describe two distinct characteristics of each type. **6**

b) Context switching is required when program control is re-directed to a different address location temporarily. What might cause this re-direction? **1**

c) When context switching takes place, the current program context must be saved. Describe what is meant by the program context. **3**

Total 10 Marks

© NCC Education Limited 2025

Question 10

a) Briefly describe ONE (1) advantage of a multi-core **and** ONE (1) advantage of a quantum computing systems. **3**

State which of these systems might be better suited for running accounting software.

b) When creating software, programmers must be considerate of utilising system resources efficiently. In what TWO (2) ways can the compiler help with optimisation of code? **2**

c) Describe TWO (2) benefits of using distributed computer systems. **2**

d) Explain three key characteristics of embedded systems and how they impact their design and performance. **3**

Total 10 Marks

End of paper

© NCC Education Limited 2025